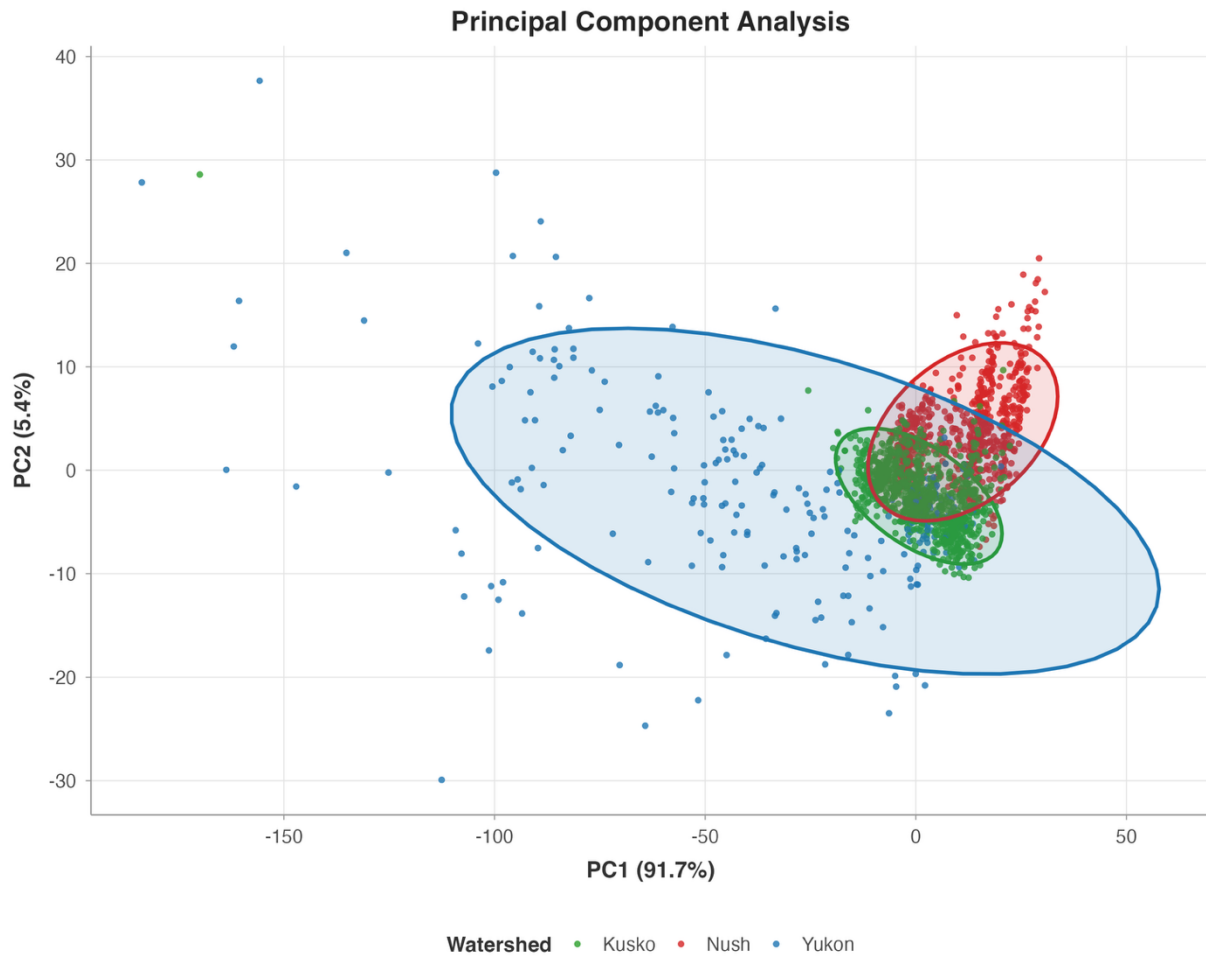
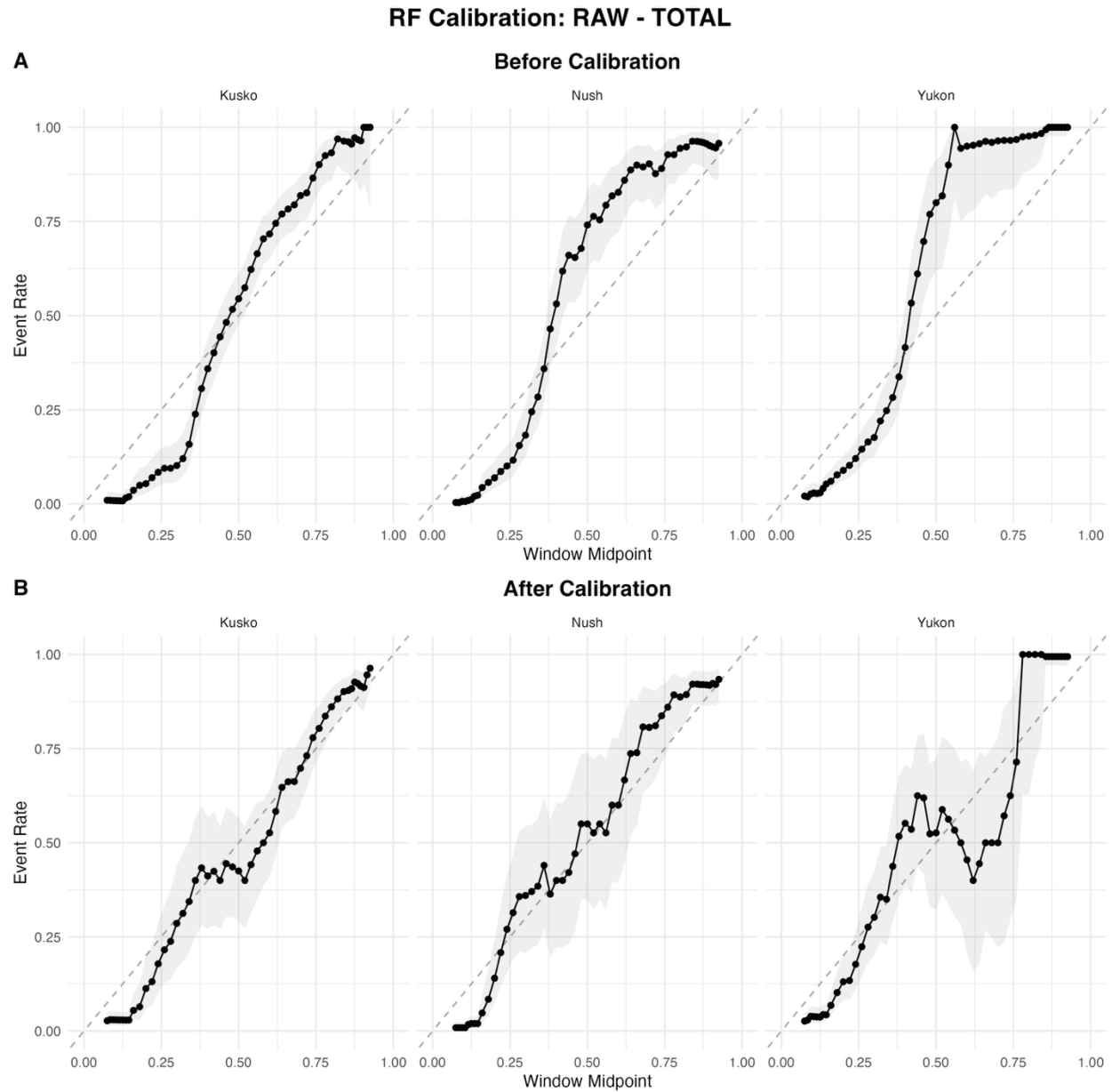




Supp Figure 1: PCA of GAM smoothed data with all Middle and Upper Yukon individuals removed, leaving only CWAK individuals. Data show a similar pattern to the PCA which included all individuals, but with many less individuals in the extreme range of both PC1 and PC2. This pattern illustrates the unique shape of the timeseries for many non-CWAK individuals compared to those in the lower watershed, which are more easily distinguished.



Supp Figure 2: Example diagrams showing the relationship between predicted probability value on the x axis and true class likelihood based on event rate on the y axis for random forest prediction on the total testing set. Predicted probability scores from the model (panel A) show a relatively linear relationship, indicating that probability scores are relatively well aligned before calibration. Applying a multinomial isotonic regression recalibrates probability scores to better match true class likelihood (panel B) , which is further confirmed by an increase in the brier score

